

Organic Chemistry: Core Reaction Types

1. Substitution Reactions

One atom or group is replaced by another. SN1 proceeds via a carbocation intermediate (favored by tertiary substrates, polar protic solvents); SN2 is a single concerted step (favored by primary substrates, strong nucleophiles).

2. Addition Reactions

Two atoms/groups add across a multiple bond, converting it to a single bond — typical of alkenes and alkynes reacting with H₂, X₂, or HX.

3. Elimination Reactions

Atoms are removed from adjacent carbons to form a double bond (E1 via carbocation, E2 concerted with a strong base).

4. Quick Comparison

- SN1/E1: carbocation intermediate, favors tertiary substrates.
- SN2/E2: concerted, favors primary substrates and strong nucleophiles/bases.
- Steric hindrance is the deciding factor between substitution and elimination pathways.